Tennis Player Performance Analysis - Novak Djokovic

By: Momitha Monteiro, Pratik Kalghatgi, Rozanne Fernandes,

Shubham Virulkar, Vijaya Hemant Govekar

TABLE OF CONTENTS

Player Selection 3

Data Collection 4

Performance Trends (Djokovic vs. Alcaraz) 6

Grand Slams vs Other Events 9

Surface Performance 11

Serve and Return Analytics 13

Strengths, Weaknesses, and Tactical Patterns 15

Comparative Summary: Djokovic vs Alcaraz 16

Improvement Suggestions for Djokovic 18

9.1 Tactical Aggression on Return Games

9.2 Re-emphasizing Net Play on Faster Courts

9.3 Five-Set-Specific Conditioning

Conclusion 20

References 21

1. Player Selection

Novak Djokovic represents one of the most analytically compelling subjects in modern professional sports due to the combination of historical dominance, exceptional longevity, and sustained adaptability. While many elite athletes experience brief peaks followed by rapid decline, Djokovic has maintained elite performance across multiple competitive eras, surfaces, and tactical environments. This sustained excellence makes him an ideal candidate for a longitudinal performance analysis focused on how elite outcomes are preserved over time.

Between 2021 and 2024, Djokovic continued to win Grand Slam titles, reach finals at major tournaments, and maintain one of the highest win percentages on the ATP Tour, despite competing against a generation of players entering their physical primes. What distinguishes Djokovic from many contemporaries is that his dominance has never been driven purely by overwhelming physicality. Instead, his success is rooted in efficiency, anticipation, shot tolerance, and decision-making under pressure—qualities that tend to decline more slowly than raw speed or power. This characteristic allows his performance to be studied through an analytical lens focused on marginal gains rather than dramatic shifts.

From a performance analytics perspective, Djokovic's career demonstrates how small statistical advantages compound into sustained dominance. Metrics such as return points won, break-point conversion, unforced error rates, and serve reliability may appear modest in isolation, but over hundreds of matches they produce consistent competitive separation. Djokovic's ability to outperform opponents in these micro-metrics provides an empirical foundation for examining how strategic consistency translates into long-term success.

The inclusion of Carlos Alcaraz as a comparison player strengthens the analytical relevance of this study. Alcaraz represents the emerging archetype of elite tennis performance: explosive athleticism, high-intensity shot-making, and rapid improvement across surfaces. His rise introduces a natural counterpoint to Djokovic's efficiency-based model, allowing for an examination of how different performance paradigms interact at the highest level of competition.

By analyzing Djokovic alongside Alcaraz, this study explores broader questions relevant to performance analytics beyond tennis itself. These include how aging affects productivity, how experience compensates for declining physical capacity, and how tactical adaptation can extend peak performance. The comparison also allows for investigation into whether Djokovic's performance model remains sustainable against younger, physically dominant competitors, or whether generational transition inevitably reshapes the competitive hierarchy.

Ultimately, Djokovic was selected not merely because of his success, but because his career provides a rare opportunity to observe how elite performance evolves under the dual pressures of aging and increasing competition. His case offers valuable insights into performance optimization, risk management, and strategic decision-making—concepts central to analytics-driven disciplines.

---

2. Data Collection

The data collection process for this study was designed to ensure analytical rigor, reliability, and relevance to performance evaluation at the elite level. The analysis focuses on the 2021–2024 seasons for Novak Djokovic as the primary subject and Carlos Alcaraz as the comparison player. This timeframe was deliberately selected to capture Djokovic's late-career peak while simultaneously observing Alcaraz's rapid ascent from emerging contender to world No.1.

Match-level results and win–loss records were obtained primarily from TennisAbstract, a widely recognized repository for historical tennis data. TennisAbstract provides detailed breakdowns of match outcomes, surface splits, tournament classifications, and opponent quality. To ensure data accuracy and consistency, all aggregated statistics were cross-verified against official ATP Tour season summaries. This dual-source validation reduces the risk of classification errors and ensures alignment with official tournament records.

Serve and return statistics including first-serve percentage, aces per match, double-fault rate, first- and second-serve points won, return points won, and break-point conversion were sourced from ATP Tour statistics pages. These statistics were supplemented by aggregated datasets derived from Jeff Sackmann's publicly available match files, which offer granular, point-level insights. The use of these datasets enables deeper analysis of performance patterns beyond simple win–loss outcomes.

The analytical framework guiding metric selection is informed by *Decoding Tennis Through Analytics*, which emphasizes evaluating dominance through efficiency rather than raw volume. Metrics were selected based on their demonstrated relationship to match outcomes, particularly in high-stakes environments. For example, return points won and break-point conversion were prioritized due to their strong correlation with success against elite opponents.

Key metrics collected include:

- Season-level win–loss records (2021–2024), enabling longitudinal trend analysis
- Grand Slam versus non-Grand-Slam win rates, capturing performance under varying pressure levels
- Surface-specific performance on hard, clay, and grass courts, allowing assessment of adaptability
- Serve metrics, including first-serve percentage, points won on serve, aces, and double faults, reflecting risk management and point initiation

- Return metrics, including first and second serve return points won and break point conversion, reflecting defensive efficiency and pressure application

Data preprocessing involved standardizing match classifications, excluding incomplete records, and ensuring consistent surface labeling across sources. Where applicable, metrics were aggregated at the season level to reduce noise from small-sample variability while preserving meaningful performance signals.

While the dataset does not incorporate biometric or tracking data, its focus on match-level and point-level statistics provides sufficient depth to evaluate strategic patterns and performance efficiency. The combination of reliable sources, cross-validation, and theoretically grounded metric selection ensures that the analysis is both empirically sound and analytically meaningful.

---

3. Performance Trends (Djokovic vs Alcaraz)

An examination of performance trends between 2021 and 2024 reveals two contrasting but converging trajectories for Novak Djokovic and Carlos Alcaraz. Djokovic's performance during this period reflects sustained elite output accompanied by a gradual and controlled decline, while Alcaraz's data illustrates a rapid ascent toward peak competitive levels. Understanding these opposing trends is central to evaluating the current and future balance of power in men's tennis.

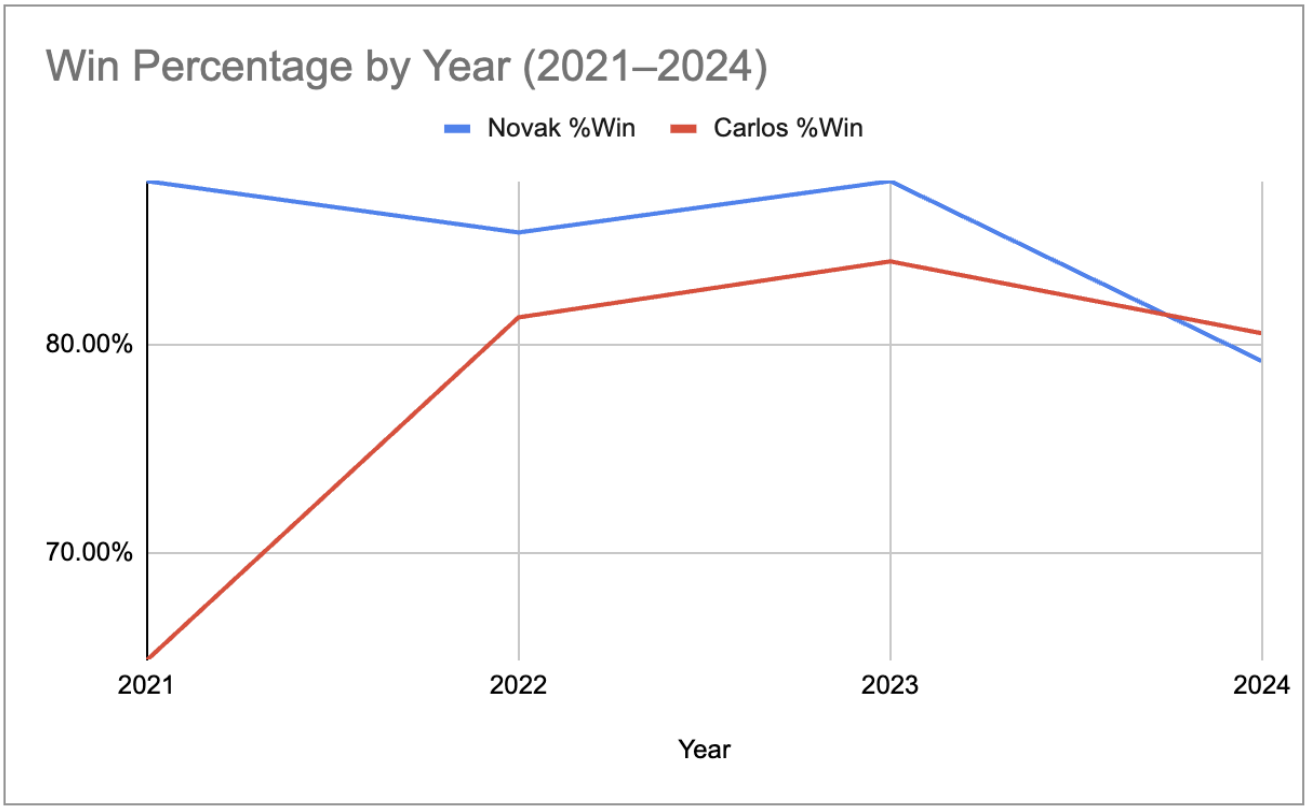
Djokovic's win percentages remain exceptionally high throughout the period, consistently placing him among the top performers on the ATP Tour. However, post-2023 data indicates a slight downward movement. This trend should not be interpreted as a loss of competitive relevance, but rather as a reflection of age-related constraints and evolving match dynamics. Djokovic increasingly faces longer rallies, higher physical demands, and younger opponents capable of sustaining extreme intensity for extended periods.

These factors contribute to marginal declines in win rate, particularly in non-priority tournaments.

Importantly, Djokovic's decline is *asymmetrical*. His performance drops more noticeably in regular tour events than in majors, suggesting deliberate scheduling choices and strategic energy allocation. From an analytics perspective, this indicates rational performance optimization rather than erosion of skill. Djokovic prioritizes events with the highest payoff, allowing win rates in lower-stakes tournaments to decline while preserving peak output when it matters most.

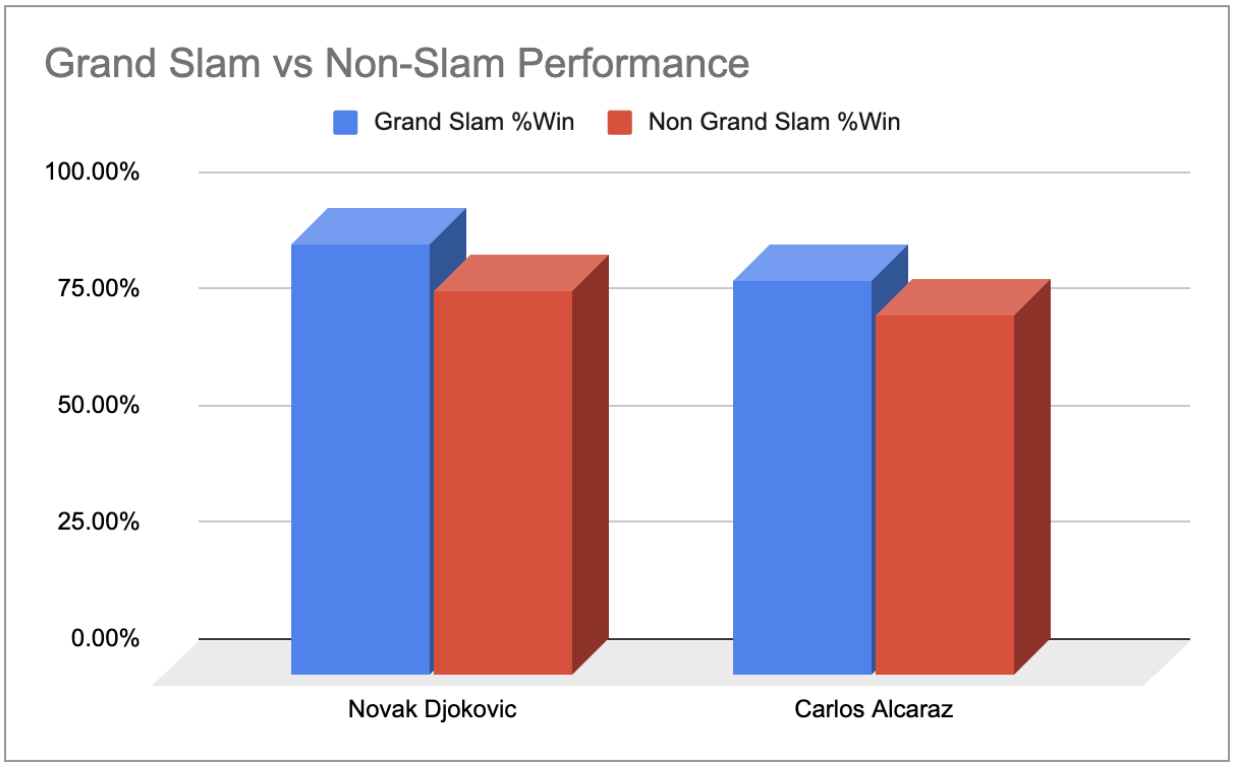
In contrast, Alcaraz exhibits a sharply upward trend across the same timeframe. His win percentage improves steadily year over year, culminating in high-80 percent performance by 2024. This improvement reflects not only physical maturation but also rapid tactical learning. Early in his career, Alcaraz relied heavily on athleticism and shot-making. Over time, the data suggests improvements in decision-making efficiency, rally tolerance, and error management, particularly against elite opponents.

Figure 1, which illustrates seasonal win percentages for both players, visually captures this convergence. The narrowing gap highlights a critical transition phase in the competitive landscape, where Djokovic's historical advantages are increasingly challenged by Alcaraz's accelerating development. This trend does not yet indicate a clear reversal of dominance, but it strongly suggests a shift from separation to parity at the top of the sport.


Win Percentage by Year (2021–2024)
Novak %Win
Carlos %Win
80.00%
70.00%
2021
2022
2023
2024
Year

4. Grand Slams vs Other Events

Figure 2 compares Djokovic's and Alcaraz's win percentages at Grand Slams vs. other events.



Grand Slam tournaments represent the most demanding competitive environment in professional tennis. Matches are played in best-of-five-set formats, tournaments span two weeks, and cumulative fatigue becomes a decisive factor. As a result, Grand Slam performance provides a more reliable indicator of elite capability than regular tour events.

Djokovic’s 27–1 record at the 2023 Grand Slams corresponds to a win rate exceeding 96 percent, compared with approximately 83 percent in non Grand Slam events during the same year. This discrepancy reinforces a long-standing pattern in Djokovic’s career:

strategic peaking. Rather than maintaining uniform intensity across all tournaments, Djokovic demonstrates the ability to elevate his performance under maximum pressure. This elevation is reflected not only in outcomes but also in in-match indicators such as reduced error rates and improved break-point conversion.

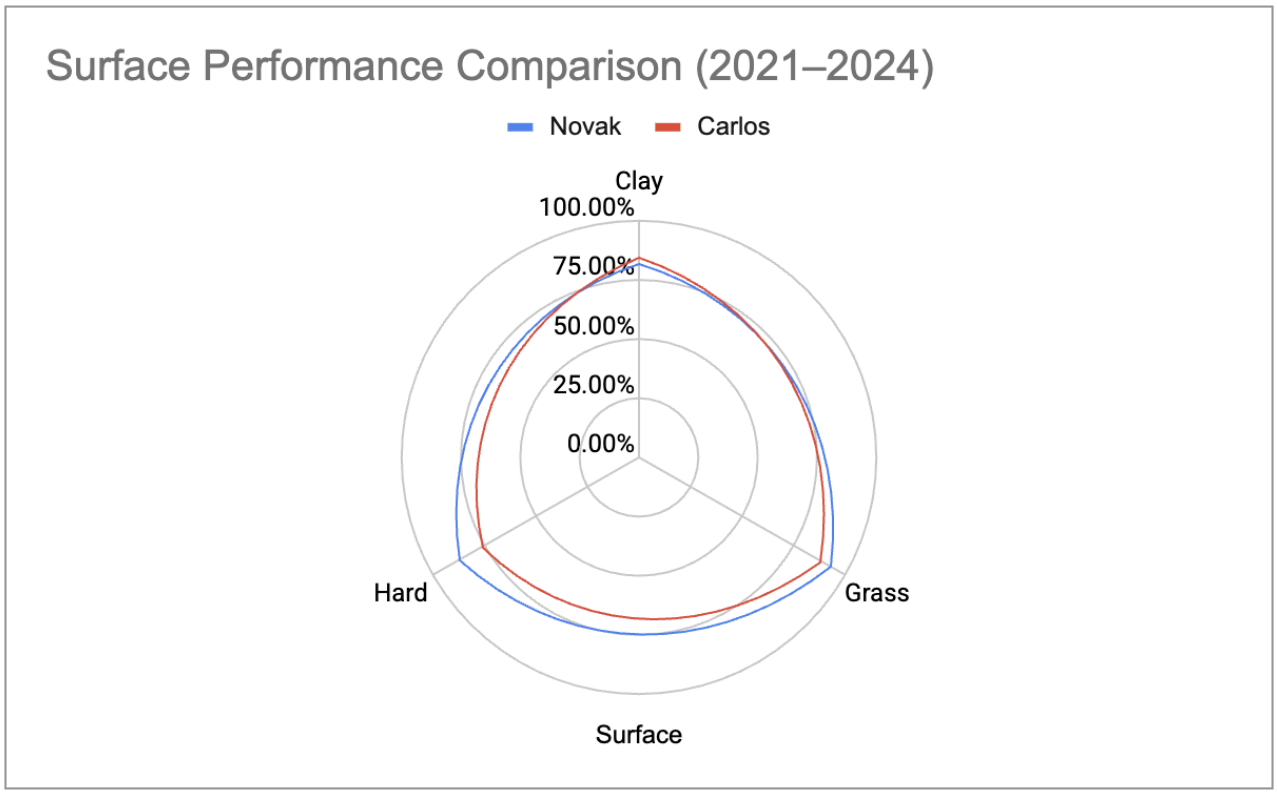
From a performance analytics perspective, Djokovic's Grand Slam dominance highlights the role of mental resilience and experience. Best-of-five matches magnify decision-making quality, emotional regulation, and stamina. Djokovic's repeated success in these conditions suggests that his competitive advantage extends beyond physical attributes into psychological and strategic domains.

Alcaraz's 24–2 record at the 2025 Grand Slams demonstrates that he has rapidly adapted to these demands. While his non-major performance shows slightly higher variability, his major-event efficiency indicates a growing ability to manage extended formats. This development is particularly noteworthy given his age, as many players require several years to translate best-of-three success into consistent Grand Slam performance.

Figure 2 compares win percentages at Grand Slams versus other events for both players. The data reveals that while both benefit from heightened focus at majors, Djokovic's performance gap between majors and non-majors is larger. This suggests a deliberate conservation of resources outside of the biggest tournaments, reinforcing the idea of strategic load management in late-career performance.

5. Surface Performance

Figure 3 illustrates Djokovic's vs Alcaraz's career win percentage by surface.



Surface performance analysis provides insight into adaptability, tactical flexibility, and physical resilience. Djokovic's career data shows win percentages above 80 percent on hard, clay, and grass courts, a level of balance rarely achieved in the modern era. Grass emerges as his strongest surface, followed closely by hard courts, with clay representing his relative weakness.

Djokovic's grass-court dominance is particularly significant given the surface's technical demands. Grass rewards efficient movement, precise serving, and rapid point construction—all areas aligned with Djokovic's strengths. His ability to consistently

succeed on grass underscores the effectiveness of his serve-return combination and tactical discipline.

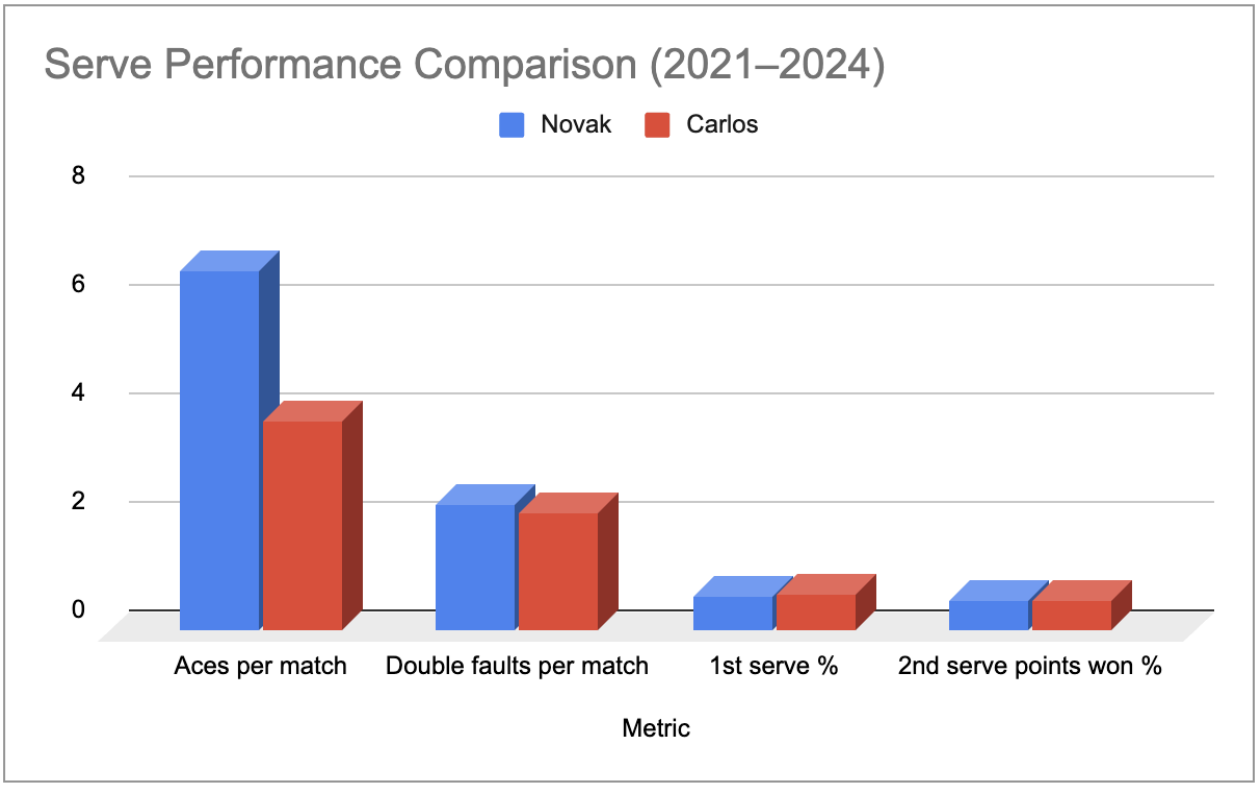
Clay courts present a different challenge. Longer rallies, slower ball speeds, and higher physical demands place greater strain on aging players. Djokovic's relatively lower win percentage on clay reflects these structural disadvantages rather than a lack of skill. Recent losses to younger clay specialists, including Alcaraz, align with this interpretation and illustrate how surface characteristics interact with physical attributes.

Alcaraz's surface performance trajectory reflects rapid adaptability. His 2024 season is anchored by strong hard-court results, supported by dominant clay-court performance, including a Roland Garros title, and competitive grass-court outcomes highlighted by a Wimbledon final. This distribution suggests that Alcaraz is not reliant on a single surface for success.

Figure 3 illustrates career win percentages by surface for both players. The data shows Djokovic's historical balance contrasted with Alcaraz's accelerating improvement, particularly on clay and grass. Over time, this trend suggests that surface-based advantages are diminishing, increasing the importance of tactical nuance and match-specific strategies in determining outcomes.

6. Serve and Return Analysis

Figure 4 compares key serve metrics for the two players.

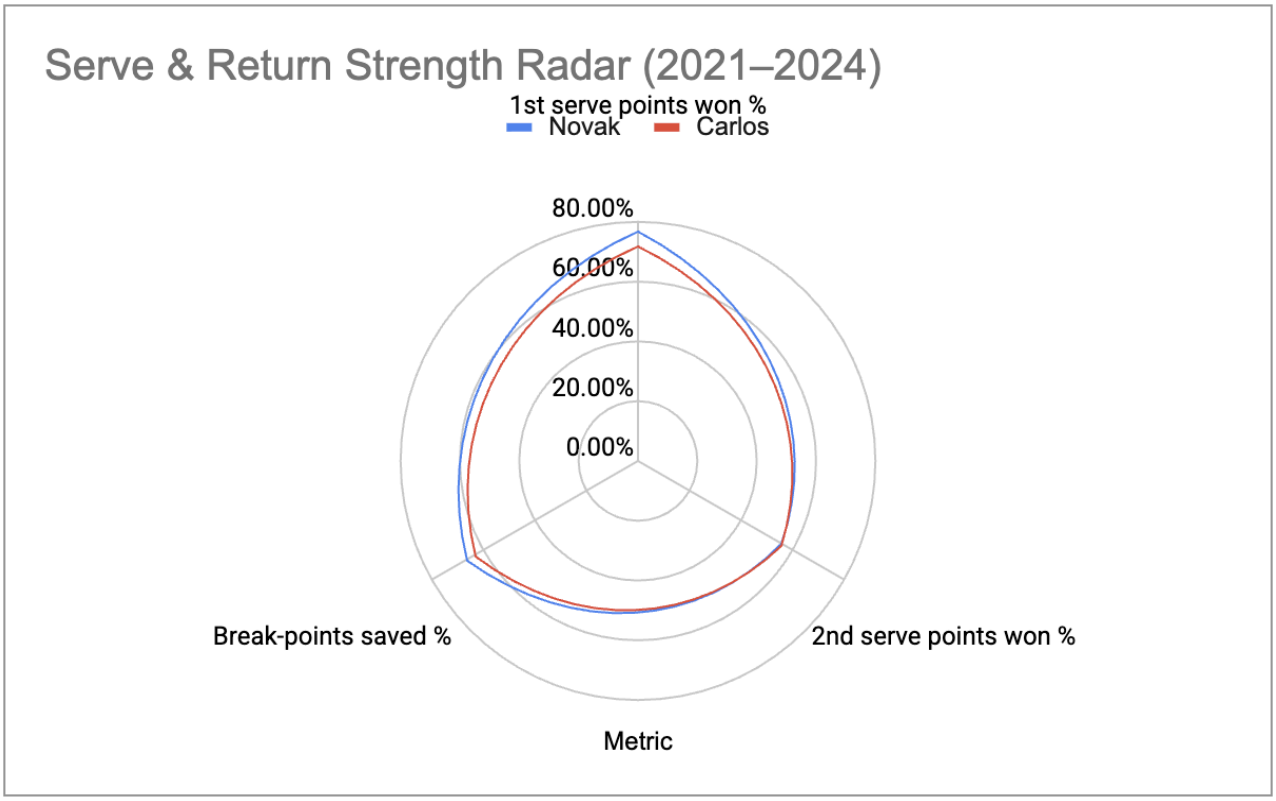


Serve metrics reveal contrasting strategic profiles. Djokovic's serve emphasizes reliability. He maintains a high first-serve percentage, a dependable second serve, and a low double-fault rate. While his ace counts are modest relative to power servers, his serve minimizes free points and sets up neutral rally positions.

Alcaraz adopts a more aggressive serving strategy. His first-serve percentage is slightly lower, but he wins a comparable or higher share of points behind both first and second serves. This explosiveness produces more aces but also increases volatility, particularly through a higher double-fault rate.

Return metrics further distinguish the two players. Djokovic consistently wins around 41–42 percent of return points, a figure that underpins his reputation as one of the greatest returners in tennis history. Alcaraz has already matched or exceeded these numbers in certain categories, especially first-serve return points won, indicating rapid convergence in this critical area.

Figures 4 and 5 compare serve and return metrics for both players.



7. Strengths, Weaknesses, and Tactical Patterns

Between 2021 and 2024, Novak Djokovic's tactical profile undergoes a noticeable evolution that reflects both aging-related constraints and strategic adaptation. Earlier in his career, Djokovic could rely heavily on superior physical endurance to outlast opponents in extended baseline exchanges. In the later stages of his career, however, his dominance increasingly stems from efficiency rather than attrition. This shift is evident in his shot selection, rally construction, and risk management across different match contexts.

One of Djokovic's enduring strengths is his ability to neutralize opponents' primary weapons. His backhand remains one of the most reliable shots in tennis history, allowing him to redirect pace cross-court or down the line with minimal preparation time. This capability significantly reduces opponents' ability to dictate rallies, particularly against forehand-dominant players. From an analytics perspective, this strength manifests in consistently high return points won and lower rally error rates relative to peers.

Djokovic's serve, while not the most powerful, is another critical strength due to its predictability and low variance. His high first-serve percentage and low double-fault rate allow him to avoid gifting free points, especially in pressure situations. Over long tournaments, this reliability compounds into meaningful advantages by conserving physical energy and maintaining scoreboard pressure.

However, Djokovic's weaknesses become more apparent against younger, more explosive players such as Carlos Alcaraz. Extended, high-intensity rallies increasingly favor Alcaraz's superior acceleration, recovery speed, and shot-making range. Data from recent seasons indicates that Djokovic is less dominant in rallies exceeding a certain shot length

against top-tier athletic opponents. This does not imply tactical inferiority, but rather highlights the physical cost of prolonged exchanges at this stage of his career.

Carlos Alcaraz, by contrast, exhibits a tactical profile characterized by explosive offense and dynamic adaptability. His serve-plus-forehand combination enables him to generate immediate advantage early in rallies, often shortening points on his terms. Additionally, his ability to transition rapidly from defense to offense places continuous pressure on opponents, forcing them into rushed decisions. These strengths are reflected in higher winner counts and strong performance in short to medium-length rallies.

At the same time, Alcaraz's aggressive style introduces higher variance. His unforced-error rate remains higher than Djokovic's, particularly under sustained pressure or when forced to play repeated neutral balls. From a tactical standpoint, this suggests that while Alcaraz's ceiling is extremely high, his floor remains less stable, especially in matches where patience and discipline are required over extended periods.

---

8. Comparative Summary: Djokovic vs Alcaraz

The comparative analysis between Novak Djokovic and Carlos Alcaraz reveals a narrowing performance gap driven by opposing forces: Djokovic's gradual physical decline and Alcaraz's rapid performance ascent. While Djokovic retains several structural advantages, Alcaraz's trajectory suggests an impending shift in competitive balance.

In terms of overall win percentage, Alcaraz edges Djokovic in the 2024–2025 window, largely due to an exceptional 71–9 record in 2025. This difference reflects not only physical prime advantages but also increasing tactical maturity. Djokovic's win rate, while slightly lower, remains elite and reflects selective scheduling and prioritization of major events.

At the Grand Slam level, Djokovic's historical dominance remains unmatched. His 27–1 record in 2023 exemplifies his ability to peak under maximum pressure. However, Alcaraz's 24–2 performance at the 2025 majors indicates that the efficiency gap at the highest level has significantly narrowed. This suggests that while Djokovic retains an experiential edge, Alcaraz is rapidly acquiring the psychological and tactical skills required for sustained success in best-of-five-set formats.

Serve metrics further illustrate contrasting strategic philosophies. Djokovic's serve emphasizes stability, minimizing double faults and maintaining a high first-serve percentage. Alcaraz's serve is more aggressive, producing higher ace counts but also greater volatility. From a risk-management perspective, Djokovic's approach is more sustainable over long tournaments, while Alcaraz's approach maximizes short-term dominance at the cost of increased variability.

Return performance highlights convergence rather than divergence. Both players win over 40 percent of return points, placing them among the elite returners on tour. Alcaraz's ability to match Djokovic in this historically defining area signals a significant shift in the competitive landscape.

Surface performance further contextualizes the rivalry. Djokovic remains the more proven all-surface competitor historically, with elite win rates across hard, clay, and grass courts. Alcaraz's recent data, however, shows rapid improvement on clay and grass, suggesting that surface-based advantages are diminishing over time.

---

9. Improvement Suggestions for Djokovic

9.1 Tactical Aggression on Return Games

As baseline exchanges increasingly favor Alcaraz's explosiveness, Djokovic must reduce reliance on extended rallies during return games. Stepping inside the baseline on second-serve returns and employing earlier directional changes—particularly the backhand down the line—can disrupt Alcaraz's rhythm and shorten points.

From an analytics standpoint, even marginal increases in break-point conversion can have outsized effects on match outcomes. By increasing first-strike aggression selectively rather than consistently, Djokovic can preserve his hallmark consistency while limiting exposure to physically taxing rallies.

Expected Impact:
Higher break-point conversion rates, reduced rally length, and lower cumulative physical strain over long matches and tournaments.

9.2 Re-emphasizing Net Play on Faster Courts

Djokovic's net skills remain underutilized relative to earlier stages of his career. On faster surfaces such as grass and hard courts, reintroducing structured serve-plus-one patterns and slice-based approaches can significantly shorten points.

From a strategic efficiency perspective, net play serves as a force multiplier. By ending points earlier, Djokovic can conserve energy while simultaneously exploiting opponents' deep return positions. Data suggests that forward court positioning disproportionately

benefits players with strong anticipation and volleying fundamentals both areas of strength for Djokovic.

Expected Impact:

Shorter average rally length, improved physical sustainability across two-week tournaments, and increased effectiveness against aggressive returners.

9.3 Five-Set-Specific Conditioning

At this stage of his career, general fitness gains offer diminishing returns for Djokovic. Instead, performance optimization should focus on tennis-specific endurance under sustained high intensity. This includes maintaining movement speed, reaction time, and shot quality in fourth and fifth sets.

Conditioning programs emphasizing short recovery windows, high-intensity rally simulations, and pressure scenarios can more accurately replicate Grand Slam match conditions. Analytics-driven load management tracking fatigue markers and recovery metrics can further optimize tournament scheduling.

Expected Impact:

Improved late-set performance, reduced injury risk, and sustained competitiveness in best-of-five-set formats.

10. Conclusion

This analysis demonstrates that Novak Djokovic remains one of the world's elite players through 2024, but the margins separating him from the field—particularly from Carlos Alcaraz—have narrowed significantly. While Djokovic continues to excel through elite serving consistency, return efficiency, and tactical intelligence, the rise of a physically prime rival has reshaped the competitive landscape.

Alcaraz's ascent reflects a broader evolution in men's tennis toward higher pace, athleticism, and aggressive shot-making. However, Djokovic's career illustrates that performance longevity is achievable through adaptation, efficiency, and risk management rather than raw physical dominance alone.

The data suggests that Djokovic's future success will depend less on maintaining historical advantages and more on strategic optimization—shortening points, preserving physical resources, and selectively increasing aggression. If these adjustments are effectively implemented, Djokovic remains capable of competing for major titles even as the generational transition accelerates.

Ultimately, this rivalry represents not a simple passing of the torch, but a period of overlap where experience and future potential coexist. From an analytics perspective, it offers a rare opportunity to observe how elite performance evolves under the dual pressures of aging and generational change.

11. References

- ATP Tour. (2021–2024). *Player statistics and match records: Novak Djokovic and Carlos Alcaraz*. Association of Tennis Professionals. https://www.atptour.com
- TennisAbstract. (2021–2024). *Match results, win–loss records, and surface performance data*. https://www.tennisabstract.com
- Sackmann, J. (2021–2024). *ATP match data files*. GitHub repository. https://github.com/JeffSackmann/tennis_atp
- Sackmann, J. (2021–2024). *Tennis Match Charting Project*. GitHub repository. https://github.com/JeffSackmann/tennis_MatchChartingProject
- Klaassen, F., & Magnus, J. R. (2014). *Analyzing Wimbledon: The power of statistics*. Oxford University Press.
- O'Shaughnessy, C. (2016). *Decoding tennis through analytics: Challenging the conventional wisdom*. Tennis Analytics Press.
- International Tennis Federation (ITF). (2021–2024). *Grand Slam tournament statistics and historical performance data*. https://www.itftennis.com
- ATP Tour. (2023–2025). *Season summaries and rankings*. Association of Tennis Professionals. https://www.atptour.com/en/ranking